



Smart Medicine Reminder System for DVAGO

Software Test Report

Submitted by

Abdul Ghani (2606-2022)

Syed Anas Ali (2640-2022)

Supervisor

Mr. Muhammad Salman

External Supervisor

Mr. Muhammad Hamza

Bachelor of Science in Computer Science 2026

Faculty of Engineering Sciences and Technology

Hamdard University, Main Campus, Karachi, Pakistan

Software Test Plan

Project Title: Smart Medicine Reminder System for DVAGO

Date: 11-Jun-2026

Version: 1.0

Test Plan

The following table presents the detailed test plan, including the schedule for each functional component of the DVAGO platform.

S.No	Description	Tested by	Start Date	End Date
1	Login and Sign-Up	Abdul Ghani	06-Jun-2026	06-Jun-2026
2	Onboarding & Notification Permission	Abdul Ghani	07-Jun-2026	07-Jun-2026
3	Dashboard – Next Dose & Mark as Taken	Abdul Ghani	08-Jun-2026	08-Jun-2026
4	Add Medicine (Manual Entry)	Abdul Ghani	10-Jun-2026	10-Jun-2026
5	Scan Medicine (AI Wrapper Scan)	Abdul Ghani	10-Jun-2026	10-Jun-2026
6	My Medicines – List, Search & Filters	Abdul Ghani	11-Jun-2026	11-Jun-2026
7	Dose Reminder Notification (Lock Screen)	Abdul Ghani	11-Jun-2026	11-Jun-2026
8	Notifications Panel & Activity Log	Abdul Ghani	11-Jun-2026	11-Jun-2026
9	Analytics Dashboard (Adherence Tracking)	Abdul Ghani	11-Jun-2026	11-Jun-2026
10	Profile Management & Logout	Abdul Ghani	11-Jun-2026	11-Jun-2026

Test Cases

Test Case 1 – Login, Sign-Up & Network Retry

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Login and Sign-Up

Date: 06-Jun-2026

Test Case ID: TC_001

Test Engineer: Abdul Ghani

Test Case Description: Testing user login with valid credentials, and verifying the Sign-Up tab is accessible.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Open app and reach Login screen	–	DVAGO splash loads; Login / Sign Up tabs displayed	Login screen loaded with Login/Sign Up tabs	Pass
2	Enter email & password, tap Login	davidyen.agc@gmail.com / 123456	User authenticated and redirected to Dashboard	Login succeeded, Dashboard loaded	Pass
3	Tap “Sign Up” tab	–	Sign-Up form displayed	Sign-Up tab switched correctly	Pass
4	Check “Continue with Google” option	–	Google sign-in option visible on both tabs	Google button visible and tappable	Pass

Test Case 2 – Onboarding & Notification Permission

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Onboarding & Notification Permission

Date: 07-Jun-2026

Test Case ID: TC_002

Test Engineer: Abdul Ghani

Test Case Description: Verify first-launch notification permission prompt and onboarding carousel.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Launch app for the first time	–	System dialog requests notification permission	“Allow DVAGO-SMR to send you notifications?” dialog shown	Pass
2	Tap “Allow”	–	Permission granted; app proceeds to onboarding	Permission granted, onboarding screen displayed	Pass
3	Swipe through onboarding slides	–	Slides “Never Miss a Dose” and “Track Your	Both slides rendered correctly with pagination dots	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
			Health” shown with Skip/Next controls		
4	Tap “Next” / final slide action	—	Onboarding completes and Login screen is shown	Redirected to Login screen	Pass

Test Case 3 – Dashboard – Next Dose & Mark as Taken

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Dashboard – Next Dose & Mark as Taken

Date: 08-Jun-2026

Test Case ID: TC_003

Test Engineer: Abdul Ghani

Test Case Description: Confirm the dashboard shows the correct upcoming dose, live countdown, and supports marking a dose as taken.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Open Dashboard after login	—	Greeting banner, date/time, and summary cards (Medicines, Adherence, Today) load	“Good Evening, Welcome back!” banner with 6 Medicines / 100% Adherence / 3 Today shown	Pass
2	View “Next Dose” card	—	Card shows medicine name, dosage, scheduled time and live countdown	“CalPol 200mg, 11:49 PM, Time Remaining 24h 15m 6s” displayed and ticking down	Pass
3	Tap “Mark as Taken” when due	Dose due (timer at 0s)	Dose recorded as taken; next dose in queue loads	Card updated and next scheduled medicine populated	Pass
4	Review “Today’s Schedule” list	—	All medicines due today are listed with status ticks	CalPol, Motilium, Panadol Syrup, Panadol Night and Motilium 2 listed with green checkmarks	Pass
5	Use “Quick Actions” (Add Medicine / Scan Wrapper / History)	—	All three shortcuts are visible and tappable from Dashboard	All three quick-action tiles rendered and functional	Pass

Test Case 4 – Add Medicine (Manual Entry)

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Add Medicine (Manual Entry)

Date: 10-Jun-2026

Test Case ID: TC_004

Test Engineer: Abdul Ghani

Test Case Description: Verify a new medicine can be added manually with dosage, frequency, reminder times, duration, stock and instructions.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Tap “Add Medicine”	–	Add Medicine form opens	Form opened with Medicine Name & Dosage fields	Pass
2	Enter medicine name & dosage	Panadol Night, 500	Fields accept text/numeric input	“Panadol Night” and “500” entered successfully	Pass
3	Select frequency	4 Times Daily	Frequency option list expands and selection is saved	“4 Times Daily” selected	Pass
4	Add reminder time(s)	11:37 AM (via “+ Add Time”)	Time picker opens; selected time added to Reminder Times list	11:37 AM added to Reminder Times	Pass
5	Enter Duration, Stock Quantity & Instructions	1 day, Qty 2, “Before Sleep”	All fields accept input and are retained	1 / 2 / “Before Sleep” saved in form	Pass
6	Tap “Add Medicine” (submit)	–	Success confirmation shown; medicine saved and notification reminder note displayed	“Success – Medicine added successfully!” dialog shown	Pass
7	Verify new medicine on Dashboard / My Medicines	–	“Panadol Night 500 mg” appears as Next Dose / in medicine list with live countdown	Card shown with countdown timer ticking down correctly	Pass

Test Case 5 – Scan Medicine (AI Wrapper Scan)

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Scan Medicine (AI Wrapper Scan)

Date: 10-Jun-2026

Test Case ID: TC_005

Test Engineer: Abdul Ghani

Test Case Description: Validate the camera-based medicine wrapper scan auto-detects medicine details and pre-fills the Add Medicine form.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Tap “Scan Wrapper” / “Scan Medicine”	–	Scan Medicine screen opens with framing guide and instructions	Instructions shown: “Ensure good lighting,	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
				Hold phone steady, Capture label clearly”	
2	Tap “Scan Medicine” to open camera	—	Camera view opens for capture	Camera opened in Photo mode	Pass
3	Capture medicine wrapper	Motilium (Domperidone) box	Photo captured and processed by AI	Photo captured successfully	Pass
4	Review detection result	—	“Medicine Detected!” confirmation with auto-filled Name & Dosage	“Medicine Detected!” shown; Name=“Motilium”, Dosage=“120” pre-filled	Pass
5	Complete remaining fields (Category, Frequency, Duration, Stock, Instructions, Reminder Time)	Vomiting / Once / 1 day / Qty 1 / “Before Meal” / 11:40 PM	All fields editable and accept manual input after auto-fill	All fields updated and retained correctly	Pass
6	Tap “Save Medicine”	—	Success confirmation shown; medicine saved to list	“Success – Medicine saved successfully!” shown; “Motilium 2” added	Pass

Test Case 6 – My Medicines – List, Search & Filters

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: My Medicines – List, Search & Filters

Date: 11-Jun-2026

Test Case ID: TC_006

Test Engineer: Abdul Ghani

Test Case Description: Confirm the My Medicines screen lists all medicines and supports search and status filters (All / Active / Low Stock).

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Navigate to “Medicines” tab	—	Full medicine list loads with search bar and filter chips	List loaded with “All / Active / Low Stock” chips and counts	Pass
2	Tap “All” filter	—	All medicines displayed regardless of stock status	All 8 medicines listed	Pass
3	Tap “Active” filter	—	Only medicines with an active schedule are shown	Active-only subset displayed	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
4	Tap “Low Stock” filter	–	Only medicines below stock threshold are shown	Low Stock subset (2 items) displayed	Pass
5	Use search bar	“Panadol”	List narrows to matching medicine names only	Filtered results matched query	Pass
6	Open a medicine card	Panadol Syrup	Card expands to show dosage, frequency and next reminder	120ml, Once daily, Next: 09:32 shown correctly	Pass

Test Case 7 – Dose Reminder Notification (Lock Screen)

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Dose Reminder Notification (Lock Screen)

Date: 11-Jun-2026

Test Case ID: TC_007

Test Engineer: Abdul Ghani

Test Case Description: Verify the app fires a system notification and lock-screen alert at the scheduled dose time.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Lock device shortly before scheduled dose time	Scheduled time: 11:40 PM	Device remains locked until reminder fires	Lock screen displayed normally, no premature alert	Pass
2	Wait for scheduled time to elapse	–	Notification “Time to take [Medicine]” appears on lock screen	“Time to take Motilium 2 – It is time to take 120 ml. Please ta...” shown on lock screen	Pass
3	Pull down notification shade	–	Same reminder notification is visible and expandable in shade	Notification visible in shade with app icon and message	Pass
4	Unlock device and open app from notification	–	App opens directly to Dashboard showing the due dose	Dashboard opened with “Due now!” status on Next Dose card	Pass

Test Case 8 – Notifications Panel & Activity Log

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Notifications Panel & Activity Log

Date: 11-Jun-2026

Test Case ID: TC_008

Test Engineer: Abdul Ghani

Test Case Description: Confirm the in-app Notifications panel logs dose-taken and reminder events, and supports Clear All.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Tap bell icon on Dashboard	—	Notifications panel opens	Panel opened with “No notifications” initial state	Pass
2	Trigger a dose reminder and mark it as taken	—	Two log entries are created: Dosage Reminder and Dose Taken	“Dose Taken – Panadol Night 500 mg was marked as taken” and “Dosage Reminder – Motilium 2 120 ml is scheduled for 11:40 PM” both logged with timestamps	Pass
3	Re-open Notifications panel	—	Notification badge count matches number of unread entries	Badge showed “2” matching unread entries	Pass
4	Tap “Clear All”	—	All notification entries are removed and badge resets	List cleared, badge count reset to 0	Pass

Test Case 9 – Analytics Dashboard (Adherence Tracking)

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Analytics Dashboard (Adherence Tracking)

Date: 11-Jun-2026

Test Case ID: TC_009

Test Engineer: Abdul Ghani

Test Case Description: Validate the Analytics tab computes adherence statistics correctly across Week / Month / Year views and surfaces insights.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Navigate to “Analytics” tab	—	Analytics screen loads with Week/Month/Year toggle	Screen loaded; “Week” selected by default	Pass
2	View “Week” adherence	—	Bar chart and On-Time Rate / Streak stats render for the week	83% On-Time Rate, 1-day streak, week bar chart (Thu/Fri/Sat) shown	Pass
3	Switch to “Month” view	—	Stats recalculate for monthly range	86% Adherence Rate, 6/7 Doses Taken, Month Adherence chart updated	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
4	Switch to “Year” view	—	Stats recalculate for yearly range	Year view loaded (data refreshed for year range)	Pass
5	Review “Medicine Breakdown” section	—	Per-medicine adherence percentage listed (e.g., Panadol, Motilium, CalPol)	Panadol 100%, Ponstan 100%, CalPol 50%, Motilium 100%, Panadol Syrup 100% listed with progress bars	Pass
6	Review “Insights” section	—	Adherence rate, On-Time Rate, Current Streak and Best Performance Time summarised	“Your week adherence rate is 86%”, “83% of taken doses were on time”, “Current streak: 1 days”, Best Performance Time “Morning” shown	Pass

Test Case 10 – Profile Management & Logout

Project Name: Smart Medicine Reminder System for DVAGO

Iteration No: 1.0

Module Name: Profile Management & Logout

Date: 11-Jun-2026

Test Case ID: TC_010

Test Engineer: Abdul Ghani

Test Case Description: Confirm profile details can be viewed/edited and that logout requires confirmation and returns the user to the Login screen.

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
1	Navigate to “Profile” tab	—	Profile screen loads with name, email, and summary stats	“Abdul Ghani” profile loaded with Medicines/Adherence/Days Active stats	Pass
2	Tap “Edit Profile”	—	Edit dialog opens with editable Name and Email fields	Edit Profile dialog opened with pre-filled Name/Email and Cancel/Save actions	Pass
3	Open “Health Profile” sub-sections	—	Personal Information, Medical History, Prescriptions and My Doctors entries are accessible	All four sub-sections listed and tappable	Pass
4	Open “Notification Settings”	—	Toggles for Push Notifications, Sound, Vibration and Voice Reminders are shown	All four toggles displayed and switchable	Pass

S.No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
5	Tap “Logout”	—	Confirmation dialog “Are you sure you want to logout?” is shown	Dialog shown with CANCEL / LOGOUT options	Pass
6	Confirm “Logout”	—	Session ends and user is redirected to Login screen	Redirected to Login screen successfully	Pass

End of Test Plan & Test Cases